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#include <iostream> #include <cmath> using namespace std;
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void printRecursionTree(int n, int level) {  
    if (n == 1) {  
        for (int i = 0; i < level; i++) {  
            cout << " ";  
        }  
        cout << "T(1) = 1" << endl;  
        return;  
    }  
  
    for (int i = 0; i < level; i++) {  
        cout << " ";  
    }  
    cout << "T(" << n << ") = 2T(" << n/2 << ") + " << n << endl;  
  
    printRecursionTree(n / 2, level + 1);  
    printRecursionTree(n / 2, level + 1); }  
  
int main() {  
    cout << "Recursion-Tree Method Example" << endl;  
    cout << "Recurrence:  $T(n) = 2T(n/2) + n$ " << endl;  
    cout << "Solution:  $T(n) = O(n \log n)$ " << endl;  
    cout << endl;  
  
    cout << "Recursion Tree for T(8):" << endl;  
    printRecursionTree(8, 0);  
  
    cout << endl;  
    cout << "Cost at each level:" << endl;  
    cout << "Level 0:  $n = 8$ " << endl;  
    cout << "Level 1:  $n/2 + n/2 = 8$ " << endl;  
    cout << "Level 2:  $n/4 + n/4 + n/4 + n/4 = 8$ " << endl;  
    cout << "Level 3:  $1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 = 8$ " << endl;  
    cout << endl;  
    cout << "Total cost:  $n * \log(n) = 8 * 3 = 24$ " << endl;  
  
    return 0; }
```